

Midterm exam

1. When H_2SO_4 is neutralized by NaOH in aqueous solution, what is the net ionic equation?
 - A) $2\text{H}^+ + 2\text{NaOH} \rightarrow 2\text{H}_2\text{O} + 2\text{Na}^+ + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{Na}^+$
 - B) $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$
 - C) $\text{H}_2\text{SO}_4 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{SO}_4^{2-}$
 - D) $\text{SO}_4^{2-} + 2\text{Na}^+ \rightarrow \text{Na}_2\text{SO}_4$
2. What is the number of O atoms in 6.15 g of $\text{Al}(\text{PO}_4)_3$ (molar mass = 122.00 g/mol)?
 - A) 3.04×10^{22}
 - B) 7.60×10^{21}
 - C) 1.21×10^{23}
 - D) 2.40×10^{24}
3. Barium reacts with a certain element to form a compound with the general formula Ba_3X_2 . What would the most likely formula be for the compound formed between sodium and element X?
 - A) NaX_2
 - B) Na_2X
 - C) NaX_3
 - D) Na_3X
4. The precision of a balance is ± 0.001 . How many significant figures are in an object with a mass of 26 g?
 - A) 5
 - B) 3
 - C) 4
 - D) 2
5. A certain organic compound is composed of C, H, and O. Decomposition of 5.00 g of this compound produces 12.397 g of CO_2 and 4.437 g of H_2O . What is the empirical formula of this compound?
 - A) $\text{C}_4\text{H}_2\text{O}$

- B) C_4H_4O
- C) C_2H_5O
- D) C_3H_2O

6. Of the three types of radioactivity characterized by Rutherford, which is/are not particles?

- A) β -rays
- B) α -rays, β -rays, and γ -rays
- C) γ -rays
- D) α -rays and β -rays

7. Which of the following is TRUE:

- A) Infrared radiation has a higher frequency than UV radiation.
- B) Gamma ray radiation has a higher wavelength than visible radiation.
- C) The energy of 0.50 mole of photons = $3.01 \times 10^{23} \times (h\nu / C) 3.01 \times 10^{23} \times (h\nu / C)$.
- D) The energy of 100 photons equals $102 (h C / \lambda) 102 (h C / \lambda)$.

8. Which of the following are weak electrolytes?

- A) NaOH
- B) HF
- C) $NaNO_3$
- D) NH_3

9. Which of the following is not a valid set of four quantum numbers? (n, l, ml, mss)

- A) 2, 0, 0, +1/2
- B) 2, 1, 0, -1/2
- C) 1, 0, 0, +1/2
- D) 1, 1, 0, +1/2

10. Which one of the following reactions is a combination reaction?

- A) $2 NH_3 (g) \rightarrow N_2 (g) + 3 H_2 (g)$
- B) $C (s) + O_2 (g) \rightarrow CO_2 (g)$
- C) $HCl (aq) + NaOH (aq) \rightarrow NaCl (aq) + H_2O$
- D) $CH_4 (g) + 2 O_2 (g) \rightarrow CO_2 (g) + 2 H_2O$

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	B	B	A	B	C	C	B, D	D	B

11. The number of C atoms in 4.50 g of $\text{C}_2\text{H}_2\text{O}_4$ (molar mass = 90 g/mol) is:

- ☐ A) 1.20×10^{23}
- ☐ B) 4.60×10^{21}
- ☐ C) 6.02×10^{22}
- ☐ D) 1.50×10^{22}

12. Which of the following are strong electrolytes?

- ☐ A) HCl
- ☐ B) $\text{HC}_2\text{H}_3\text{O}_2$
- ☐ C) NH_3
- ☐ D) KCl

13. Given the following reaction: $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$. If the reaction of 0.55 mol of NH_3 with 2.0 mol of O_2 yields 10.6 g of H_2O , what is the percentage yield of H_2O (molar mass = 18.0 g/mol)?

- ☐ A) 32%
- ☐ B) 71%
- ☐ C) 24%
- ☐ D) 82%

14. The energy (J) required for an electronic transition in a Bohr hydrogen atom from $n=2$ to $n=3$ is:

- ☐ A) 3.03×10^{-19}
- ☐ B) 8.71×10^{-20}
- ☐ C) 4.02×10^{-19}
- ☐ D) 3.00×10^{-18}

15. Which one of the following statements is False (not correct)?

- ☐ A) The limiting reactant is that reactant which is consumed first.
- ☐ B) Density is an intensive physical property.
- ☐ C) The percentage composition of sodium (Na) in the two samples of 15.6 g of Na_2CO_3 and 54.2 g of Na_2CO_3 are the same.
- ☐ D) 0.5 mol of each of the following compounds; K_2S and CS_2 have the same mass of sulfur (S).

16. The temperature of a winter day is 34.0°C . What is the temperature in Fahrenheit scale ($^\circ\text{F}$)?

- ☐ A) 93.2
- ☐ B) 29.2
- ☐ C) 50.1
- ☐ D) 7.5

17. Do the following calculation and give the answer to the correct number of significant figures:

$$4.6 \div (42.6 + 8.84) \times 4.6 \div (42.6 + 8.84)$$

- ☐ A) 11.182
- ☐ B) 11
- ☐ C) 11.17
- ☐ D) 11.2

18. The element X has three naturally occurring isotopes. The masses (amu) and % abundances of the isotopes are given in the table below. The average atomic mass of the element is _____ amu.

Mass	Abundance	Isotope
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15.33	28.60	^{15}X
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17.26	13.30	^{17}X
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18.11	58.10	^{16}X
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- ☐ A) 17.20
- ☐ B) 17.11
- ☐ C) 17.65
- ☐ D) 16.90

19. Which one of the following reactions is a combustion reaction?

- A) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$
- B) $\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g}) + \text{HCl}(\text{g})$
- C) $\text{Al}_2\text{O}_3(\text{s}) + 3\text{C}(\text{s}) \rightarrow 2\text{Al}(\text{s}) + 3\text{CO}(\text{g})$
- D) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

20. The temperature of a winter day is 28.0 °F. What is the temperature in Celsius scale (°C)?

- A) 82.4
- B) -136.8
- C) -2.2
- D) 33.3

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
C	A, D	B	A	D	A	D	B	A	C

21. Which one of the following reactions is a neutralization reaction?

- A) $2\text{RbCl}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{RbClO}_3(\text{s})$
- B) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- C) $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{l})$
- D) $\text{CH}_4(\text{g}) + \text{Br}_2(\text{g}) \rightarrow \text{C}_6\text{H}_5\text{Br}(\text{g}) + \text{HBr}(\text{g})$

22. The energy (J) required for an electronic transition in a Bohr hydrogen atom from $n=2$ to $n=4$ is:

- A) $5.45 \times 10^{-19}\text{J}$
- B) $8.71 \times 10^{-20}\text{J}$
- C) $4.00 \times 10^{-19}\text{J}$
- D) $4.09 \times 10^{-19}\text{J}$

23. The ground state electron configuration of Cr is:

- A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$

- B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
- C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$
- D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^3$

24. Calculate the frequency of an electron moving at 3.66×10^6 m/s. (Mass of electron = 9.11×10^{-28} g)

- A) 1.51×10^{18} s⁻¹
- B) 5.98×10^{17} s⁻¹
- C) 1.50×10^{21} s⁻¹
- D) 1.09×10^{17} s⁻¹

25. An atom of the most common isotope of copper, ^{63}Cu , has _____ protons, _____ neutrons, and _____ electrons.

- A) 29, 29, 63
- B) 63, 29, 63
- C) 34, 29, 34
- D) 29, 34, 29

26. The ground state electron configuration of Mo is:

- A) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^4 5s^2$
- B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 5p^6 5s^2 4d^4$
- C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^4$
- D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1 4d^5$

27. In which species does sulfur have the highest oxidation number?

- A) HSO_3^-
- B) H_2S
- C) SO_2
- D) NaSO_4^-

28. The element _____ is the most similar to bromine (Br) in chemical and physical properties.

- A) At
- B) Ba
- C) Rb
- D) Li

29. Which one of the following reactions is a combustion reaction?

- A) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$
- B) $\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g}) + \text{HCl}(\text{g})$
- C) $\text{Al}_2\text{O}_3(\text{s}) + 3\text{C}(\text{s}) \rightarrow 2\text{Al}(\text{s}) + 3\text{CO}(\text{g})$
- D) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

30. The temperature of a winter day is 28.0 °F. What is the temperature in Celsius scale (°C)?

- A) 82.4
- B) -136.8
- C) -2.2
- D) 33.3

Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
B	A	B	C	D	D	D	A	A	C

31. Which one of the following reactions is a neutralization reaction?

- A) $2\text{RbCl}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{RbClO}_3(\text{s})$
- B) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- C) $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$
- D) $\text{CH}_4(\text{g}) + \text{Br}_2(\text{g}) \rightarrow \text{C}_6\text{H}_5\text{Br}(\text{g}) + \text{HBr}(\text{g})$

32. The energy (J) required for an electronic transition in a Bohr hydrogen atom from $n=2$ to $n=4$ is:

- A) $5.45 \times 10^{-19}\text{J}$
- B) $8.71 \times 10^{-20}\text{J}$
- C) $4.00 \times 10^{-19}\text{J}$
- D) $4.09 \times 10^{-19}\text{J}$

33. The ground state electron configuration of Cr is:

- ☐ A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
- ☐ B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
- ☐ C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$
- ☐ D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^3$

34. Calculate the frequency of an electron moving at 3.66×10^6 m/s. (Mass of electron = 9.11×10^{-28} g)

- ☐ A) 1.51×10^{18} s⁻¹
- ☐ B) 5.98×10^{17} s⁻¹
- ☐ C) 1.50×10^{21} s⁻¹
- ☐ D) 1.09×10^{17} s⁻¹

35. An atom of the most common isotope of copper, ^{63}Cu , has _____ protons, _____ neutrons, and _____ electrons.

- ☐ A) 29, 29, 63
- ☐ B) 63, 29, 63
- ☐ C) 34, 29, 34
- ☐ D) 29, 34, 29

36. The ground state electron configuration of Mo is:

- ☐ A) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^4 5s^2$
- ☐ B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 5p^6 5s^2 4d^4$
- ☐ C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^4$
- ☐ D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1 4d^5$

37. In which species does sulfur have the highest oxidation number?

- ☐ A) HSO_3^-
- ☐ B) H_2S
- ☐ C) SO_2
- ☐ D) NaSO_4^-

38. The element _____ is the most similar to bromine (Br) in chemical and physical properties.

- ☐ A) At

- B) Ba
- C) Rb
- D) Li

39. Which one of the following reactions is a combustion reaction?

- A) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$
- B) $\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g}) + \text{HCl}(\text{g})$
- C) $\text{Al}_2\text{O}_3(\text{s}) + 3\text{C}(\text{s}) \rightarrow 2\text{Al}(\text{s}) + 3\text{CO}(\text{g})$
- D) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

40. The temperature of a winter day is 28.0 °F. What is the temperature in Celsius scale (°C)?

- A) 82.4
- B) -136.8
- C) -2.2
- D) 33.3

Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40
B	A	B	C	D	D	D	A	A	C

41. Which one of the following reactions is a neutralization reaction?

- A) $2\text{RbCl}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{RbClO}_3(\text{s})$
- B) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- C) $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{l})$
- D) $\text{CH}_4(\text{g}) + \text{Br}_2(\text{l}) \rightarrow \text{C}_6\text{H}_5\text{Br}(\text{l}) + \text{HBr}(\text{g})$

42. The energy (J) required for an electronic transition in a Bohr hydrogen atom from $n=2$ to $n=4$ is:

- A) 5.45×10^{-19}
- B) 8.71×10^{-20}
- C) 4.00×10^{-19}

- D) 4.09×10^{-19}

43. The ground state electron configuration of Cr is:

- A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
- B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
- C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$
- D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^3$

44. Calculate the frequency of an electron moving at 3.66×10^6 m/s. (Mass of electron = 9.11×10^{-28} g)

- A) 1.51×10^{18} s⁻¹
- B) 5.98×10^{17} s⁻¹
- C) 1.50×10^{21} s⁻¹
- D) 1.09×10^{17} s⁻¹

45. An atom of the most common isotope of copper, ^{63}Cu , has _____ protons, _____ neutrons, and _____ electrons.

- A) 29, 29, 63
- B) 63, 29, 63
- C) 34, 29, 34
- D) 29, 34, 29

46. The ground state electron configuration of Mo is:

- A) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^4 5s^2$
- B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 5p^6 5s^2 4d^4$
- C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^4$
- D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1 4d^5$

47. In which species does sulfur have the highest oxidation number?

- A) HSO_3^-
- B) H_2S
- C) SO_2

- D) NaSO_4^-

48. The element _____ is the most similar to bromine (Br) in chemical and physical properties.

- A) At
- B) Ba
- C) Rb
- D) Li

49. Which one of the following reactions is a combustion reaction?

- A) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$
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- C) $\text{Al}_2\text{O}_3(\text{s}) + 3\text{C}(\text{s}) \rightarrow 2\text{Al}(\text{s}) + 3\text{CO}(\text{g})$
- D) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

50. The temperature of a winter day is 28.0 °F. What is the temperature in Celsius scale (°C)?

- A) 82.4
- B) -136.8
- C) -2.2
- D) 33.3

Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50
B	A	B	C	D	D	D	A	A	C

51. Which one of the following reactions is a neutralization reaction?

- A) $2\text{RbCl}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{RbClO}_3$
- B) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}$
- C) $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$
- D) $\text{CH}_4(\text{g}) + \text{Br}_2(\text{g}) \rightarrow \text{C}_6\text{H}_5\text{Br}(\text{g}) + \text{HBr}(\text{g})$

52. The energy (J) required for an electronic transition in a Bohr hydrogen atom from $n=2$ to $n=4$ is:

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- B) $8.71 \times 10^{-20} \text{10}^{-20}$
- C) $4.00 \times 10^{-19} \text{10}^{-19}$
- D) $4.09 \times 10^{-19} \text{10}^{-19}$

53. The ground state electron configuration of Cr is:

- A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
- B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
- C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$
- D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^3$

54. Calculate the frequency of an electron moving at $3.66 \times 10^6 \text{ m/s}$. (Mass of electron = $9.11 \times 10^{-28} \text{ g}$)

- A) $1.51 \times 10^{18} \text{ s}^{-1}$
- B) $5.98 \times 10^{17} \text{ s}^{-1}$
- C) $1.50 \times 10^{21} \text{ s}^{-1}$
- D) $1.09 \times 10^{17} \text{ s}^{-1}$

55. An atom of the most common isotope of copper, ^{63}Cu , has _____ protons, _____ neutrons, and _____ electrons.

- A) 29, 29, 63
- B) 63, 29, 63
- C) 34, 29, 34
- D) 29, 34, 29

56. The ground state electron configuration of Mo is:

- A) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^4 5s^2$
- B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 5p^6 5s^2 4d^4$
- C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^4$
- D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1 4d^5$

57. In which species does sulfur have the highest oxidation number?

- A) HSO_3^-
- B) H_2S
- C) SO_2
- D) NaSO_4^-

58. The element _____ is the most similar to bromine (Br) in chemical and physical properties.

- A) At
- B) Ba
- C) Rb
- D) Li

59. Which one of the following reactions is a combustion reaction?

- A) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$
- B) $\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g}) + \text{HCl}(\text{g})$
- C) $\text{Al}_2\text{O}_3(\text{s}) + 3\text{C}(\text{s}) \rightarrow 2\text{Al}(\text{s}) + 3\text{CO}(\text{g})$
- D) $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

60. The temperature of a winter day is 28.0 °F. What is the temperature in Celsius scale (°C)?

- A) 82.4
- B) -136.8
- C) -2.2
- D) 33.3

Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60
B	A	B	C	D	D	D	A	A	C